Reframing the Triangle: From Pascal to von Neumann

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August 17, 2026

Abstract

This is an investigation of a triangular array of numbers from a simple arithmetical procedure. Using Pascal's triangle as a reference, the investigation of a foundational triangle of numbers is coined after John von Neumann.

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1 Introduction

Pascal's triangle will be reviewed so that the reader will be familiar with the triangular array before proceeding to the von Neumann triangle. The von Neumann triangle is a triangular array of numbers (in the form of Pascal's triangle) in which the entries are sums of two complementary monotonic sequences.

The von Neumann triangle should be familiar to the reader but in a different form than the one presented here. The triangle is named after John von Neumann who defined natural numbers using set theory. The triangle is a sieve of primes, a multiplication table, and exhibits many interesting properties as related to the natural numbers. The triangle is referenced in OEIS A158823 as well as other sequences that are derived from the triangle.

1.1 Purpose of This Study

This paper is written to be:

- A reference to Pascal's triangle
- A point of reference for the von Neumann triangle
- Possibility of finding new combinatorial aspects of both triangles

2 Pascal's Triangle

2.1 Overview

Pascal's triangle is a triangular array of numbers in which the entries are the binomial coefficients. Each number is the sum of the two numbers directly above it. The triangle is named after Blaise Pascal, though it was known to mathematicians in earlier centuries, particularly in the work of Omar Khayyám and other Persian and Chinese mathematicians.

2.2 Construction

Pascal's triangle is constructed by placing 1 at the top, and then filling each subsequent row such that each entry is the sum of the two entries above it:

```
          1
        1   1
      1   2   1
    1   3   3   1
  1   4   6   4   1
1   5  10  10   5   1
1   6  15  20  15   6   1
```

2.3 Mathematical Formula

The entries in Pascal's triangle can be expressed using the binomial coefficient:

$$T(n,k) = \binom{n}{k} = \frac{n!}{k!(n-k)!} \tag{1}$$

where n is the row number (starting from 0) and k is the position in the row (starting from 0).

2.4 Properties and Applications

Some key properties of Pascal's triangle include:

- Each row sum equals 2^n, where n is the row number
- The entries provide coefficients in the binomial expansion: $(a+b)^n = \sum_{k=0}^{n} \binom{n}{k} a^{n-k} b^k$
- The diagonal sums produce Fibonacci numbers
- It exhibits numerous symmetries and patterns useful in combinatorics and probability

3 Von Neumann's Triangle

3.1 Overview

Von Neumann's triangle is a triangular array of numbers (in the form of Pascal's triangle) in which the entries are

$$n(n_{\max} - n + 1) \tag{2}$$

for each row 1 to $n_{\max}$. The triangle is named after Von Neumann who defined natural numbers using set theory.

3.2 Construction

Begin with row 1 which is just:

$$1$$

Row 2 is composed of:

$$1 + 1$$

$$2$$

Row 3:

$$1+1+1$$

$$2+2$$

$$3$$

Row 4:

$$1+1+1+1$$

$$2+2+2$$

$$3+3$$

$$4$$

First row entry is 1.
The second row entries are $1+1$ and 2.
The third row entries are $1+1+1$, $2+2$ and 3.
Fourth row entries are $1+1+1+1$, $2+2+2$, $3+3$ and 4.
And so on, until the last row:

$$1(n_{\max}-1+1)$$

$$2(n_{\max}-2+1)$$

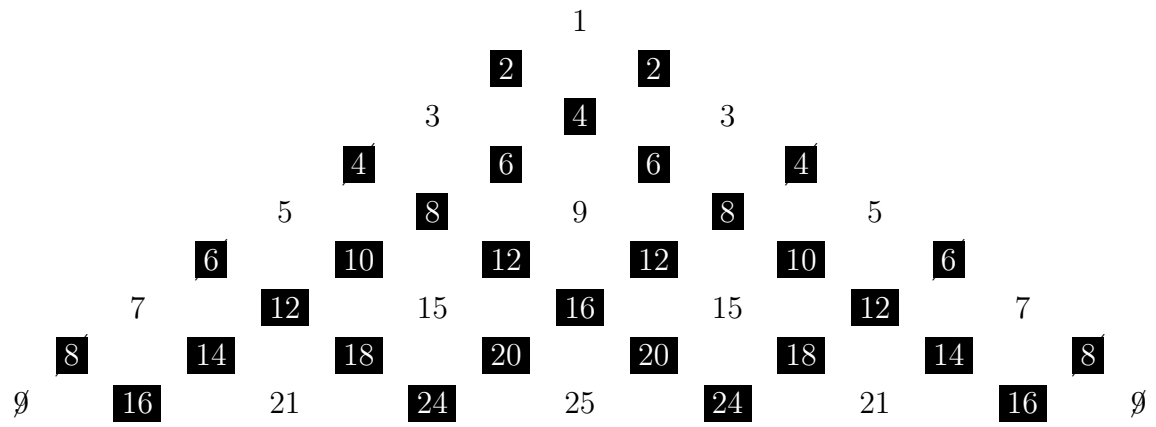
$$3(n_{\max}-3+1)$$

$$\dots$$

$$n_{\max}(n_{\max}-n_{\max}+1)$$

Von Neumann's triangle is constructed by placing 1 at the top, and then filling each subsequent row by the construction above. The resulting triangular array is referenced in OEIS A158823.

$$\begin{array}{ccccccccccccccccccc}
 & & & & & & & 1 & & & & & & & & & & & \\
 & & & & & & 2 & & 2 & & & & & & & & & & \\
 & & & & & 3 & & 4 & & 3 & & & & & & & & & \\
 & & & & 4 & & 6 & & 6 & & 4 & & & & & & & & \\
 & & & 5 & & 8 & & 9 & & 8 & & 5 & & & & & & & \\
 & & 6 & & 10 & & 12 & & 12 & & 10 & & 6 & & & & & & \\
 & 7 & & 12 & & 15 & & 16 & & 15 & & 12 & & 7 & & & & & \\
 8 & & 14 & & 18 & & 20 & & 20 & & 18 & & 14 & & 8 & & & & \\
 9 & & 16 & & 21 & & 24 & & 25 & & 24 & & 21 & & 16 & & 9 & &
\end{array}$$



3.3 Mathematical Formula

The entries in the von Neumann triangle can be expressed using the formula:

$$n_{ij} = j(i - j + 1); i = 1 \ldots n_{\max}, j <= i \tag{3}$$

where i is the row number (starting from 1), and j is the position in the row (starting from 1).

3.4 Properties and Applications

Some key properties of the Von Neumann triangle include:

- The outside diagonal entries are the natural numbers $1, 2, 3, \ldots$
- The triangle is a sieve of primes - removing the outer diagonal entries that exist in the inner triangle leaves only the prime numbers
- The triangle is a multiplication table - the entries are the products of two outer numbers intersecting on their two diagonals
- The sum of each row is exhibited in the OEIS sequence A000292
- The central vertical entries are the squares of the natural numbers $1^2, 2^2, 3^2, 4^2 \ldots$, which are referenced by OEIS sequence A000290
- The sequence of the vertical entries 2, 6, 12, 20, 30, 42, 56, 72, 90, 110, 132, 156, 182, 210, 240, 272, 306, 342, 380, 420 is the OEIS sequence A002378
- The sequence of the vertical entries 3, 8, 15, 24, 35, 48, 63, 80, 99, 120, 143, 168, 195, 224, 255, 288, 323, 360, 399 is the OEIS sequence A005563

- Vertical sequence entries 4, 10, 18, 28, 40, 54, 70, 88, 108, 130, 154, 180, 208, 238, 270, 304, 340, 378 is the OEIS sequence A028552
- Vertical sequence 5, 12, 21, 32, 45, 60, 77, 96, 117, 140, 165, 192, 221, 252, 285, 320, 357 is the OEIS sequence A028347
- Vertical sequence 6, 14, 24, 36, 50, 66, 84, 104, 126, 150, 176, 204, 234, 266, 300 is the OEIS sequence A028557
- Vertical sequence 7, 16, 27, 40, 55, 72, 91, 112, 135, 160, 187, 216, 247, 280 is the OEIS sequence A085158
- Vertical sequence 8, 18, 30, 44, 60, 78, 98, 120, 144, 170, 198, 228, 260 is the OEIS sequence A028563
- Vertical sequence 9, 20, 33, 48, 65, 84, 105, 128, 153, 180, 209, 240 is the OEIS sequence A028566
- Triangular numbers $a(n) = \frac{n(n+1)}{2}$ is the OEIS sequence A000217 and is also found in the path in the triangle starting at the top, moving repeatedly two steps out diagonally and one step in diagonally.

4 Conclusion

A quick review of Pascal's triangle and introducing the von Neumann triangle provides some clarity about how natural number constructions work. The von Neumann triangle is a triangular array of numbers in which the entries are sums of two complementary monotonic sequences. The triangle is a sieve of primes, a multiplication table, and exhibits many interesting properties as related to the natural numbers.